RAG KNOWLEDGE BASE · RAW TEXT DUMP

Muhammad Ahmad Malik

Portfolio website — complete textual content extracted from the codebase

Source: D:\Ahmadmalikportfolio2 (Next.js 16 · React 19 · TypeScript · Tailwind CSS 4)
Primary content file: src/lib/content.ts · Public repo: github.com/AHMADMALIK1376/Ahmadmalikportfolio2
Method: strict extraction, no summarising, no paraphrasing. Text is copied verbatim from the source. Sections with no matching content are marked NOT FOUND.

Contents

1. Identity & Contact
2. About Me / Bio
3. Skills & Tech Stack
4. Projects & Experience
5. Education & Certifications
6. Services / Offerings
7. Testimonials
8. Blog / Articles
9. Navigation & UI Text
10. Any Other Text

Formatting convention for RAG ingestion: every fact is a "Field: value" line under a clear heading; list items are bullets; text inside quotation context is verbatim from the source.

1. IDENTITY & CONTACT

Full Name: Muhammad Ahmad Malik

First name (as split in the hero heading): Muhammad

Last name (as split in the hero heading): Ahmad Malik

Short name: M. Ahmad Malik

Professional Title / Role: Full-Stack AI Engineer

Rotating titles (typewriter in the hero): Full-Stack AI Engineer | MERN Developer | Cloud Architect

Tagline (hero line): I build production web apps and LLM-powered systems, and make complex AI feel simple to use.

Email Address: ahmadmalik1376@gmail.com

Phone Number: **NOT FOUND**

Phone number: the source file states it is deliberately left out of the public site ("the phone number, which does not belong on a public page"). Therefore: NOT FOUND.

Location: Rawalpindi, Pakistan

Time zone: Asia/Karachi (shown on the site as "PKT")

GitHub: https://github.com/AHMADMALIK1376

LinkedIn: https://www.linkedin.com/in/ahmadmalik1376/

Twitter / X: **NOT FOUND**

Other social media: **NOT FOUND**

Resume (PDF file served by the site): /M-Ahmad-Malik-Resume.pdf (also reachable at /resume, which redirects to it); download file name: M-Ahmad-Malik-Resume.pdf

Current employer link: https://anmapk.com

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2. ABOUT ME / BIO

Section heading (About, section 01, kicker "about me · the spec sheet")

I turn hard AI problems into things people can use.

Full biography / profile text (three paragraphs, verbatim)

Paragraph 1: Full-Stack AI Engineer with hands-on experience shipping production-grade web apps and LLM-powered systems using React.js, Node.js, and Python/FastAPI.

Paragraph 2: Built and deployed AI features including LLM integration, computer vision, and browser automation, turning complex AI capabilities into simple, usable interfaces.

Paragraph 3: Strong background in scalable API design and database architecture (MongoDB, Oracle, PostgreSQL), with a focus on clean, modular, production-ready code.

The same text as shown on the About spec sheet (numbered callouts beside the photo)

Callout 01 — tag "ships": production web apps

Callout 01 text: Full-Stack AI Engineer with hands-on experience shipping production-grade web apps and LLM-powered systems using React.js, Node.js, and Python/FastAPI.

> **Callout 02 — tag "builds":** AI that is simple to use

> **Callout 02 text:** Built and deployed AI features including LLM integration, computer vision, and browser automation, turning complex AI capabilities into simple, usable interfaces.

> **Callout 03 — tag "designs":** APIs & data architecture

> **Callout 03 text:** Strong background in scalable API design and database architecture (MongoDB, Oracle, PostgreSQL), with a focus on clean, modular, production-ready code.

> **Callout 04 — tag "now":** Full Stack AI Engineer @ Anma Tech

> **Callout 04 text:** Remote · 6-month paid internship

Photo

> **Caption under the photo:** ahmad, mid-deploy ✶

> **Alt text:** Muhammad Ahmad Malik standing against a wall, typing on a laptop.

Rating plate (the spec-sheet panel under the photo)

> **Plate header:** rating plate model: M. Ahmad Malik · Full-Stack AI Engineer

> **building since:** 2023

> **years at it:** 3+ (computed in code as current year minus 2023)

> **projects:** 9 (counted from the project list)

> **tools:** 35+ (the toolbox has 37 distinct items, rounded down to a multiple of 5)

> **hackathons:** 2

> **education:** BS Computer Science · Iqra University — Dec 2023 – Present

> **based in:** Rawalpindi — Pakistan

Fun Facts / personal interests

> **Fun Facts:** **NOT FOUND**

> **Personal interests / hobbies:** **NOT FOUND**

Professional summary or objective statement

No separate "objective" statement exists. The professional summary is the three-paragraph profile above and the hero tagline: I build production web apps and LLM-powered systems, and make complex AI feel simple to use.

Other self-descriptive text on the site

> **Hero greeting:** hello, world — i'm

> **Footer tagline:** Full-Stack AI Engineer in Rawalpindi, Pakistan. Drawn by hand, built with Next.js.

> **Page meta description:** Muhammad Ahmad Malik, Full-Stack AI Engineer in Rawalpindi, Pakistan. I build production web apps and LLM-powered systems, and make complex AI feel simple to use.

> **Collaboration invitation (Contact section):** Got a project, a role, or an idea? Write it on the card and post it — it lands in my inbox.

3. SKILLS & TECH STACK

Section: Skills (section 04, kicker "the periodic table") — heading "My toolbox."

Skills grouped by the site's own categories (src/lib/content.ts → SKILLS)

Frontend

- HTML5
- CSS3
- JavaScript (ES6+)
- React.js
- Tailwind CSS
- Vite
- Context API

Backend

- Node.js
- Express.js
- FastAPI
- Python
- RESTful APIs
- JWT Authentication

Databases

- MongoDB
- Firebase
- Oracle 21c XE
- PostgreSQL
- SQL

AI & LLM

- Anthropic Claude API
- LLM Integration
- Prompt Engineering
- Computer Vision
- Multi-modal AI

Cloud & DevOps

- Google Cloud Platform
- AWS
- Docker
- Git/GitHub
- Vercel
- Netlify

Automation & Testing

- Playwright
- Web Scraping
- Browser Automation
- Accessibility Testing (ARIA)

Languages

- Python
- TypeScript
- JavaScript
- Java
- C++
- SQL

Every distinct tool/skill named in the Skills data, as one raw list (37 items; this list also scrolls as the dark tilted marquee tape under the table)

- HTML5
- CSS3
- JavaScript (ES6+)
- React.js
- Tailwind CSS
- Vite
- Context API
- Node.js
- Express.js
- FastAPI
- Python
- RESTful APIs
- JWT Authentication
- MongoDB
- Firebase
- Oracle 21c XE
- PostgreSQL
- SQL
- Anthropic Claude API
- LLM Integration
- Prompt Engineering
- Computer Vision
- Multi-modal AI
- Google Cloud Platform
- AWS
- Docker
- Git/GitHub
- Vercel
- Netlify

- Playwright
- Web Scraping
- Browser Automation
- Accessibility Testing (ARIA)
- TypeScript
- JavaScript
- Java
- C++

Periodic-table tiles (symbol · full name · short label shown on tile · category · grid row/column)

#	Symbol	Name	Label on tile	Category	Row	Col
1	Py	Python	Python	Languages	1	1
2	Gh	Git/GitHub	Git/GitHub	Cloud & DevOps	1	13
3	Js	JavaScript	JavaScript	Languages	2	1
4	Ts	TypeScript	TypeScript	Languages	2	2
5	Ht	HTML5	HTML5	Frontend	2	8
6	Cs	CSS3	CSS3	Frontend	2	9
7	Re	React.js	React.js	Frontend	2	10
8	Tw	Tailwind CSS	Tailwind CSS	Frontend	2	11
9	Vi	Vite	Vite	Frontend	2	12
10	Cx	Context API	Context API	Frontend	2	13
11	Ja	Java	Java	Languages	3	1
12	C+	C++	C++	Languages	3	2
13	No	Node.js	Node.js	Backend	3	8
14	Ex	Express.js	Express.js	Backend	3	9
15	Fa	FastAPI	FastAPI	Backend	3	10
16	Ra	RESTful APIs	RESTful APIs	Backend	3	11
17	Jw	JWT Authentication	JWT Auth	Backend	3	12
18	Dk	Docker	Docker	Cloud & DevOps	3	13
19	Sq	SQL	SQL	Languages	4	1
20	Mg	MongoDB	MongoDB	Databases	4	2
21	Fb	Firebase	Firebase	Databases	4	3
22	Or	Oracle 21c XE	Oracle 21c XE	Databases	4	4
23	Pg	PostgreSQL	PostgreSQL	Databases	4	5
24	Gc	Google Cloud Platform	Google Cloud	Cloud & DevOps	4	6
25	Aw	AWS	AWS	Cloud & DevOps	4	7
26	Vc	Vercel	Vercel	Cloud & DevOps	4	8
27	Nl	Netlify	Netlify	Cloud & DevOps	4	9
28	Pw	Playwright	Playwright	Automation & Testing	4	10
29	Ws	Web Scraping	Web Scraping	Automation & Testing	4	11
30	Ba	Browser Automation	Browser Automation	Automation & Testing	4	12
31	Ay	Accessibility Testing (ARIA)	Accessibility (ARIA)	Automation & Testing	4	13

32	Cl	Anthropic Claude API	Claude API	AI & LLM	6	4
33	Ll	LLM Integration	LLM Integration	AI & LLM	6	5
34	Pe	Prompt Engineering	Prompt Engineering	AI & LLM	6	6
35	Cv	Computer Vision	Computer Vision	AI & LLM	6	7
36	Mm	Multi-modal AI	Multi-modal AI	AI & LLM	6	8

Tile numbering is the order of the list above (the atomic number shown on each tile). Each tile also shows, on hover, "used in" followed by the projects whose stack contains it; tools not used in a listed project show: "in the kit — part of how I work, day to day".

Technologies named in project stacks (union, from the projects data)

- Next.js
- TypeScript
- FastAPI
- PostgreSQL
- Supabase
- TipTap
- y-sweet
- Claude API
- RAG (TF-IDF)
- Python
- DOCX Export
- Streamlit
- Claude (Anthropic)
- Adzuna Jobs API
- React.js
- Node.js
- Express.js
- Oracle 21c XE
- JWT
- Recharts
- D3.js
- Chart.js
- SheetJS
- jsPDF
- SVG
- CSS Animation
- GitHub Actions
- GitHub API
- LangGraph
- Qwen
- Alibaba Cloud
- Tailwind CSS
- Firebase
- HTML5 Canvas

- VS Code Extension API

Technologies named in service "tools" lists (union)

- Python
- FastAPI
- LangGraph
- Claude API
- Qwen
- Node.js
- PostgreSQL
- MongoDB
- Docker
- React.js
- Next.js
- TypeScript
- Tailwind CSS
- Vercel
- Express.js
- Oracle 21c XE
- Figma
- Playwright

Technologies / topics declared for search engines (JSON-LD knowsAbout)

- React
- Next.js
- TypeScript
- Node.js
- FastAPI
- Python
- Large Language Models
- Retrieval-Augmented Generation
- PostgreSQL
- MongoDB

Soft skills

No section lists soft skills as such. Soft-skill-like statements that appear in the text: "delivering assigned tasks on time within a professional team"; "Gained hands-on experience with Agile methodologies and team-based development workflows"; "Holding full ownership of architecture, API design, and deployment decisions from planning through shipping, with no team support"; "Taking projects end to end: gathering requirements, designing the database, building the frontend and backend, deploying, and writing the documentation."

4. PROJECTS & EXPERIENCE

4A. WORK EXPERIENCE (section 05 "Experience" — heading "Where I've been shipping.")

Intro text: Scroll down the road: the van drives it with you, and every place I've been opens up as it arrives.

Role 1: Full Stack AI Engineer @ Anma Tech

> **Project Title / Role:** Full Stack AI Engineer
>
> **Company / Organisation:** Anma Tech
>
> **Date / Duration:** Jul 2026 – Present
>
> **Location / Type:** Remote · 6-month paid internship
>
> **Link:** https://anmapk.com

Description / Key Responsibilities (verbatim):

- Working as the sole developer on an assigned AI-powered product, owning the frontend, backend, and AI/LLM integration layer end-to-end.
- Holding full ownership of architecture, API design, and deployment decisions from planning through shipping, with no team support.

> **Tech stack listed for this role:** NOT FOUND

Role 2: Website Development Intern @ Growstep Technologies

> **Project Title / Role:** Website Development Intern
>
> **Company / Organisation:** Growstep Technologies
>
> **Date / Duration:** Jan 2026 – Apr 2026
>
> **Location / Type:** Pakistan
>
> **Link:** NOT FOUND

Description / Key Responsibilities (verbatim):

- Completed a React.js website development internship, delivering assigned tasks on time within a professional team.
- Gained hands-on experience with Agile methodologies and team-based development workflows.

> **Tech stack listed for this role:** NOT FOUND

Role 3: UI/UX Designer @ Freelance

> **Project Title / Role:** UI/UX Designer
>
> **Company / Organisation:** Freelance
>
> **Date / Duration:** Nov 2025 – Present
>
> **Location / Type:** Remote
>
> **Link:** NOT FOUND

Description / Key Responsibilities (verbatim):

- Design user-centered interfaces and interactive prototypes using Figma for web and mobile applications.
- Create wireframes, mockups, and design systems that prioritize usability, accessibility, and modern aesthetics.

> **Tech stack listed for this role:** NOT FOUND

Role 4: Freelance Full-Stack Developer @ Freelance

> **Project Title / Role:** Freelance Full-Stack Developer
>
> **Company / Organisation:** Freelance
>
> **Date / Duration:** 2023 – Present
>
> **Location / Type:** Remote
>
> **Link:** NOT FOUND

Description / Key Responsibilities (verbatim):

- Building and delivering full-stack web applications for individual clients — management systems, dashboards and portfolio sites.
- Taking projects end to end: gathering requirements, designing the database, building the frontend and backend, deploying, and writing the documentation.

> **Tech stack listed for this role:** NOT FOUND

Source note (content.ts): the Autonomous AI Testing Platform is deliberately left out of the site because its details belong to a private product. It is therefore NOT FOUND in the site content.

4B. PROJECTS (section 03 "Selected work" — heading "Stuff I've built." — shown on a 3D revolving ring of cards)

Intro text: Drag the ring, swipe it, or let it turn, and open any card to watch it work.

Project #01: Folium

> **Project Title:** Folium
>
> **Tagline:** Collaborative document editor
>
> **Role:** Full-Stack Developer
>
> **Date / Duration:** Jul 2026 – Present
>
> **Category (filter label):** Full-stack
>
> **Tech Stack used:** Next.js, TypeScript, FastAPI, PostgreSQL, Supabase, TipTap, y-sweet
>
> **How it works (the steps its animated drawing shows):** TipTap editor → Live sync · y-sweet → FastAPI · auth → Postgres · versions

Description / Key Achievements (verbatim):

- Rebuilt a timeboxed prototype into a full product: a Next.js frontend and a FastAPI backend on PostgreSQL, with real authentication through Supabase.
- Documents stored as TipTap JSON, with sharing at permission levels, soft delete to a trash folder, and file import and export.
- Version history that snapshots as you edit, with preview and restore.
- Live collaborative editing with shared cursors over y-sweet, falling back to single-user autosave when it is not configured.

> **GitHub Repo:** https://github.com/AHMADMALIK1376/Folium
>
> **Live Demo link:** NOT FOUND

Project #02: Bid & Proposal Engine

> **Project Title:** Bid & Proposal Engine
>
> **Tagline:** AI-powered RFP and tender automation
>
> **Role:** Full Stack AI Developer

> **Date / Duration:** 2026

> **Category (filter label):** AI & LLM

> **Tech Stack used:** Claude API, RAG (TF-IDF), Python, DOCX Export

> **How it works (the steps its animated drawing shows):** Upload a tender → Claude extracts → TF-IDF matching → GO / NO-GO · DOCX

Description / Key Achievements (verbatim):

- Built an end-to-end engine automating RFP/tender responses — a process that typically consumes 60–80% of bid preparation time.
- Used the Claude API to extract requirements, evaluation criteria, and deadlines from uploaded PDF/DOCX tender documents.
- Engineered a RAG-based matching system (TF-IDF) that scores requirements against a capability library with PASS/GAP/FAIL compliance checks.
- Built a win dashboard with a GO/NO-GO decision and probability score, plus AI-generated proposal drafts in 60 seconds with DOCX export.

> **GitHub Repo:** https://github.com/AHMADMALIK1376/bid-response-engine

> **Live Demo link:** **NOT FOUND**

Project #03: HireMe Agent

> **Project Title:** HireMe Agent

> **Tagline:** AI job matching and cover letter generator

> **Role:** Full Stack AI Developer

> **Date / Duration:** 2026

> **Category (filter label):** AI & LLM

> **Tech Stack used:** Streamlit, Claude (Anthropic), Adzuna Jobs API

> **How it works (the steps its animated drawing shows):** CV → fields → Adzuna search → Top 3 letters → You review

Description / Key Achievements (verbatim):

- Built an AI agent that parses an uploaded CV (PDF/DOCX) into structured fields using Claude, then searches the Adzuna Jobs API for matching roles.
- Generates a tailored 3-paragraph cover letter for each of the top 3 postings, using only content verified from the candidate's actual CV.
- Designed with a human-in-the-loop review step — users edit and copy each letter before applying; nothing is sent automatically.

> **GitHub Repo:** https://github.com/AHMADMALIK1376/JOB-MEETS

> **Live Demo link:** **NOT FOUND**

Project #04: FocusFlow

> **Project Title:** FocusFlow

> **Tagline:** Student productivity management system

> **Role:** Full-Stack Developer

> **Date / Duration:** Feb 2026 – Present

> **Category (filter label):** Full-stack

> **Tech Stack used:** React.js, Node.js, Express.js, Oracle 21c XE, JWT

> **How it works (the steps its animated drawing shows):** Tasks · schedule → Attendance triggers → Midnight cron → Email reminders

Description / Key Achievements (verbatim):

- Built a platform combining task management, academic scheduling, attendance tracking, and automated email notifications.
- Engineered a real-time attendance engine with database triggers and a midnight cron job for absence tracking.
- Designed a fully normalized Oracle database with 17 tables, stored procedures, triggers, and a materialized view.
- Built a RESTful API with 30+ endpoints across 7 modules and 5 background cron services.

> **GitHub Repo:** **NOT FOUND**

> **Live Demo link:** **NOT FOUND**

Project #05: GraphForge

> **Project Title:** GraphForge

> **Tagline:** Universal client-side data visualization platform

> **Role:** Solo Developer

> **Date / Duration:** May 2026 – Present

> **Category (filter label):** Tools & viz

> **Tech Stack used:** React.js, Recharts, D3.js, Chart.js, SheetJS, jsPDF

> **How it works (the steps its animated drawing shows):** Import data → 69+ chart types → Registry pattern → Export client-side

Description / Key Achievements (verbatim):

- Architected a modular SPA (92+ files) with 69+ chart types across 5 analytical categories.
- Built a dual export system (PNG/SVG/PDF + CSV/JSON/Excel), fully client-side for GDPR compliance.
- Achieved zero ESLint warnings across the codebase using a scalable Registry Pattern architecture.

> **GitHub Repo:** https://github.com/AHMADMALIK1376/graphforge

> **Live Demo link:** **NOT FOUND**

Project #06: Animated GitHub Profile

> **Project Title:** Animated GitHub Profile

> **Tagline:** A README that rebuilds itself from live data

> **Role:** Solo Developer

> **Date / Duration:** May 2026 – Present

> **Category (filter label):** Tools & viz

> **Tech Stack used:** Python, SVG, CSS Animation, GitHub Actions, GitHub API

> **How it works (the steps its animated drawing shows):** Daily workflow → GitHub API → Python → SVG → Profile panels

Description / Key Achievements (verbatim):

- Generates every panel on the profile as a self-contained animated SVG in Python — no scripts and no external requests, so nothing is stripped when GitHub renders it.
- A GitHub Actions workflow rebuilds the panels every day from the GitHub API and the public contributions calendar.

- Includes a world map sized by each language's share of code and a robotic-arm skills board that adds newly detected languages on its own.

> **GitHub Repo:** https://github.com/AHMADMALIK1376/AHMADMALIK1376

> **Live Demo link:** **NOT FOUND**

Project #07: NeuraCache

> **Project Title:** NeuraCache

> **Tagline:** AI agent with persistent memory

> **Role:** Hackathon Developer

> **Date / Duration:** Jun 2026

> **Category (filter label):** AI & LLM

> **Tech Stack used:** Python, FastAPI, LangGraph, Qwen, Alibaba Cloud

> **How it works (the steps its animated drawing shows):** A message in → LangGraph state → Memory profile → Qwen replies

Description / Key Achievements (verbatim):

- Built for the Global AI Hackathon: an AI agent that remembers the people it talks to, running Qwen on Alibaba Cloud Bailian.
- Orchestrated the conversation as a LangGraph state graph that carries a per-user memory profile between turns.
- Feeds remembered details — name, preferences, last topic — back into the system prompt so the agent recalls the user in later conversations.

> **GitHub Repo:** https://github.com/AHMADMALIK1376/neuraCache

> **Live Demo link:** **NOT FOUND**

Project #08: Routine Dashboard

> **Project Title:** Routine Dashboard

> **Tagline:** Task management web application

> **Role:** Full-Stack Developer

> **Date / Duration:** Apr 2026 – Present

> **Category (filter label):** Full-stack

> **Tech Stack used:** React.js, Node.js, Tailwind CSS, Firebase

> **How it works (the steps its animated drawing shows):** Groups · tasks → Live progress → Firebase sync → Dark mode

Description / Key Achievements (verbatim):

- Built a Monday.com-inspired dashboard with task CRUD, group organization, and live progress tracking.
- Implemented a custom design system with dark mode persisted via localStorage.

> **GitHub Repo:** **NOT FOUND**

> **Live Demo link:** **NOT FOUND**

Project #09: Bugistan

> **Project Title:** Bugistan

> **Tagline:** VS Code extension for gamified debugging

> **Role:** Solo Developer

> **Date / Duration:** Apr 2026 – Present

> **Category (filter label):** Tools & viz

> **Tech Stack used:** TypeScript, HTML5 Canvas, VS Code Extension API

> **How it works (the steps its animated drawing shows):** Diagnostics → Creatures spawn → Game engine → 60 FPS canvas

Description / Key Achievements (verbatim):

- Built a VS Code extension turning code diagnostics into an interactive game with 38 creature species at 60 FPS.
- Designed a 2D game engine with creature AI and a dynamic particle system across 47+ files.

> **GitHub Repo:** https://github.com/AHMADMALIK1376/bugistan

> **Live Demo link:** **NOT FOUND**

Project cards also read, for the first stack items, the first three tools as chips and "+N" for the rest; and the buttons "details +" and "code".

4C. ALONG THE WAY (also plotted on the Experience map)

> **Internship: Website Development:** Growstep Technologies — A React.js website development internship, delivered within a professional team.

> **Workshop: GitHub Open Source Collaboration:** Iqra University Islamabad — A hands-on workshop on collaborating in open source with Git and GitHub.

> **Hackathon: CUST Hackathon 2026:** National Hackathon — A national hackathon at CUST, Islamabad: from an idea to a working build, against the clock.

> **Hackathon: Atom Camp Hackathon:** Agentic AI · 16–17 Jun 2026 — Two days of building with agentic AI at Atom Camp's hackathon.

4D. EXPERIENCE MAP STOPS (text printed on the map's ticket cards)

Map title block: THE ROAD SO FAR · 2023 → today · Rawalpindi · "not to scale". Last stop label: "you are here". End link: "next stop: your team?" (links to #contact). Pin buttons are labelled "Drive to <stop title>". Ticket kicker pattern: "stop N of 5 · Role|Education" for road stops and "along the way · Workshop|Hackathon" for landmarks. Each card ends with a round rubber stamp of three short lines:

- since, 2023, freelance
- since, dec 23, bs cs
- iqra, git, workshop
- since, nov 25, design
- jan–apr, 2026, intern
- cust, 2026, hackathon
- jun 16–17, 2026, hackathon
- jul, 2026, now

Stops in order along the map: Freelance (Role) · Iqra Uni (Education) · Git workshop (Workshop) · UI/UX (Role) · Growstep (Role) · CUST (Hackathon) · Atom Camp (Hackathon) · Anma Tech (Role).

> **Stop 2 of 5 · Education — title:** BS Computer Science

> **Stop 2 organisation:** Iqra University, Islamabad (H-9)

> **Stop 2 period:** Dec 2023 – Present

> **Stop 2 badge text:** Undergraduate · in progress

> **Stop 2 point:** Studying computer science, alongside the freelance work and everything else on this map.

> **CUST Hackathon 2026 card — period / badge:** 2026 / Hackathon
>
> **Atom Camp Hackathon card — organisation / period / badge:** Atom Camp / 16–17 Jun 2026 / Agentic AI
>
> **Workshop card — badge:** Workshop

5. EDUCATION & CERTIFICATIONS

Education

> **Degree Name:** BS Computer Science
>
> **Institution:** Iqra University, Islamabad (H-9)
>
> **Year / Duration:** Dec 2023 – Present
>
> **Note (verbatim):** Studying computer science, alongside the freelance work and everything else on this map.

Certifications

The site lists no formal certificates. It lists the following internship, workshop and hackathon credentials (data name: CERTIFICATIONS), each with its issuing/hosting body:

> **1. Kind:** Internship
>
> **Name:** Website Development
>
> **Issuing / hosting body:** Growstep Technologies
>
> **Description (verbatim):** A React.js website development internship, delivered within a professional team.
>
> **2. Kind:** Workshop
>
> **Name:** GitHub Open Source Collaboration
>
> **Issuing / hosting body:** Iqra University Islamabad
>
> **Description (verbatim):** A hands-on workshop on collaborating in open source with Git and GitHub.
>
> **3. Kind:** Hackathon
>
> **Name:** CUST Hackathon 2026
>
> **Issuing / hosting body:** National Hackathon
>
> **Description (verbatim):** A national hackathon at CUST, Islamabad: from an idea to a working build, against the clock.
>
> **4. Kind:** Hackathon
>
> **Name:** Atom Camp Hackathon
>
> **Issuing / hosting body:** Agentic AI · 16–17 Jun 2026
>
> **Description (verbatim):** Two days of building with agentic AI at Atom Camp's hackathon.
>
> **Formal certification names (e.g. Coursera, AWS, Google):** **NOT FOUND**

6. SERVICES / OFFERINGS

Section 02 "What I do" — heading "From the model to the pixel." Intro: Five kinds of work, kept in one wallet. Open a card to watch how each one goes.

Service 01: LLM model training

> **Service name:** LLM model training
>
> **Description (verbatim):** Training language models end to end: preparing the data, running forward and backward passes, and following the loss down until the model is ready to save.
>
> **Process steps ("how it goes"):** Data → tokens | Forward · backward | Loss ↓ | Checkpoints

What you get (verbatim):

- Datasets cleaned, split and tokenized, ready to train on.
- Training runs that log their loss and save checkpoints as they go.
- The finished model served behind an API, with prompts engineered around it.

> **Tools:** Python, FastAPI, LangGraph, Claude API, Qwen
>
> **Seen in (projects):** NeuraCache, Bid & Proposal Engine, HireMe Agent

Service 02: Software architecture

> **Service name:** Software architecture
>
> **Description (verbatim):** Designing systems floor by floor: clients on top, the gateway that meets them, the services that do the work and the data underneath — every connection planned before anything is built.
>
> **Process steps ("how it goes"):** Clients → gateway | Services | Event bus | Data & cache

What you get (verbatim):

- A system mapped out before it is built: clients, gateway, services and data.
- API contracts and event flows agreed up front, so parts can be built in parallel.
- Caches and queues placed where they pay for themselves.

> **Tools:** Node.js, FastAPI, PostgreSQL, MongoDB, Docker
>
> **Seen in (projects):** Folium, FocusFlow

Service 03: Web applications

> **Service name:** Web applications
>
> **Description (verbatim):** Building web apps from the database up: the data, the API over it, the interface people use — then shipped through build, test and deploy to a live server.
>
> **Process steps ("how it goes"):** Database | API | Frontend | Build · test · deploy

What you get (verbatim):

- The database, the API and the interface, built together as one product.
- A build, test and deploy pipeline, so every change ships the same way.
- A live app on a real server — not a demo on a laptop.

> **Tools:** React.js, Next.js, TypeScript, Tailwind CSS, Vercel
>
> **Seen in (projects):** Folium, Routine Dashboard, GraphForge

Service 04: Backend engineering

> **Service name:** Backend engineering

> **Description (verbatim):** The server side that holds everything up: APIs that route and check every request, a cache in front of the database, workers for jobs that can wait, and monitoring that proves it's healthy.

> **Process steps ("how it goes"):** Load balancing | APIs | Cache & database | Workers · monitoring

What you get (verbatim):

- RESTful APIs that validate every request, with JWT authentication.
- A cache in front of the database, and queries that stay fast as data grows.
- Background workers and scheduled jobs for everything that can wait.

> **Tools:** Node.js, Express.js, FastAPI, PostgreSQL, Oracle 21c XE

> **Seen in (projects):** FocusFlow, Bid & Proposal Engine

Service 05: Frontend development

> **Service name:** Frontend development

> **Description (verbatim):** Interfaces that hold up when someone looks closely: every box measured, every gap deliberate, every control reachable by keyboard and every colour readable against what's behind it.

> **Process steps ("how it goes"):** Box model | Layout & spacing | Keyboard focus | Contrast · WCAG

What you get (verbatim):

- Layouts measured to the pixel, with spacing that follows one scale.
- Every control reachable by keyboard, with a focus ring you can see.
- Colours that pass WCAG contrast, and markup a screen reader understands.

> **Tools:** React.js, TypeScript, Tailwind CSS, Figma, Playwright

> **Seen in (projects):** GraphForge, Routine Dashboard, Folium

> **Wallet pocket text:** 05 ways I build · what i do · * * * * *

> **Wallet hint (desktop):** point at the wallet · click a card

> **Wallet hint (touch):** tap a card to open it

---

7. TESTIMONIALS

NOT FOUND

No testimonial, review or client quote exists anywhere in the code.

---

8. BLOG / ARTICLES

NOT FOUND

The site has no blog or articles section.

9. NAVIGATION & UI TEXT

Top navigation bar

> **Logo wordmark:** ahmad.malik

> **Logo link accessible name:** Muhammad Ahmad Malik, back to the top

> **Main menu items (in order):** About (#about) · What I do (#services) · Work (#work) · Skills (#skills) · Experience (#experience) · Contact (#contact)

> **Nav buttons:** Resume (downloads the PDF) · Menu / Close (mobile)

> **Mobile menu:** 01 about · 02 what i do · 03 work · 04 skills · 05 experience · 06 contact · Resume · GitHub · LinkedIn

> **Skip link (keyboard):** Skip to content

Section headings (number · kicker · heading · intro)

- Hero — hello, world — i'm / Muhammad Ahmad Malik / typewriter titles: Full-Stack AI Engineer | MERN Developer | Cloud Architect
- 01 · about me · the spec sheet — I turn hard AI problems into things people can use.
- 02 · what i do — From the model to the pixel. — Five kinds of work, kept in one wallet. Open a card to watch how each one goes.
- 03 · selected work — Stuff I've built. — Drag the ring, swipe it, or let it turn, and open any card to watch it work.
- 04 · the periodic table — My toolbox.
- 05 · experience — Where I've been shipping. — Scroll down the road: the van drives it with you, and every place I've been opens up as it arrives.
- 06 · contact — Let's build something together. — Got a project, a role, or an idea? Write it on the card and post it — it lands in my inbox.

Hero

> **Buttons:** See my work (→ #work) · Say hello (→ #contact) · Resume (download)

> **Info line:** Rawalpindi, Pakistan · local time shown as "HH:MM PKT"

> **Scroll cue:** scroll (→ #about)

> **Margin notes by the factory drawing:** ideas in, apps out ;; click it to drop in an idea

Hero factory illustration labels (idea → app machine)

IDEA · IDEAS · DESIGN · BUILD · TEST · SHIP · SHIPPED ✓ · COFFEE · WAREHOUSE · DOCK · BOT-1 · BOT-2 · LIVE server panel: API-01 / DB-02 / AI-03 with status "200 OK", "p99 42ms", "CPU 31%" · UPTIME 99.98% · LIVE · apps.ahmad.dev · "N DEPLOYS" · "✓ DEPLOYED". Idea notes that drop into the hopper: FOLIUM, GRAPHFORGE, NEURACACHE, HIREME, FOCUSFLOW, BUGISTAN, BID ENGINE, ROUTINE.

Projects section UI

Card face: #NN · category (AI & LLM / Full-stack / Tools & viz) · title · tagline · first point · up to 3 tool chips and "+N" · "details +" · "code". Back of card: "ahmad .malik". Under/beside the ring: #NN · category / project title / tagline of the card at the front. Category labels: All, AI & LLM, Full-stack, Tools & viz.

Project pop-up: header "#NN · category"; buttons Previous project, Next project, Close; sub-headings "how it works", "what i did"; chips for role and period; stack list labelled "Built with"; button "Code" (GitHub link).

Services (What I do) UI

List buttons show "01 … 05", the service title, steps joined with " → " and "open ↗". Card aria-label: "<title>: open". Pop-up: "0N / 05 · what i do"; buttons Previous, Next, Close; sub-headings "how it goes", "what you get", "tools"; footer "seen in <project names>".

Skills UI

Key buttons (kinds of element): Languages, Frontend, Backend, Databases, Cloud & DevOps, Automation & Testing, AI & LLM. Featured panel: category label, tool name, and "used in <projects>" or "in the kit — part of how I work, day to day". Row label: "✶ the AI & LLM series →". Note: "point at an element to see where I've used it ↖". Aria-label for the key: "Kinds of tool".

Contact section UI (airmail postcard form + post box)

- Card greeting: Dear Ahmad,
- Message placeholder: Tell me about the project, the role, or the idea… (hidden label: "Message, at least 10 characters"; min 10, max 5000 characters)
- Sign-off: — yours, <typed name or …>
- Address block: stamp here (placeholder box) · to: Muhammad Ahmad Malik / Rawalpindi, Pakistan
- Field labels: from (placeholder: your name; max 100) · reply to (placeholder: you@example.com; max 200)
- Form accessible name: Write to Ahmad
- Send button: Send it → while sending: Sending…
- After sending (note): Sent! Thanks, <first name>. / Your letter is in the post, on its way to my inbox. I'll write back soon. / button: Write another
- Envelope: to <name> <location> · from <typed name or "a friend"> · stamp: PAKISTAN, Rs 26 · postmark ring: RAWALPINDI · PAKISTAN · <DD MON YYYY>
- Post box scene labels: POST · COLLECTION · ANY TIME ✓ · MAIL · BOT-3
- Buttons under the card: Copy email (→ "Copied" with a bubble "in your clipboard!"; hidden text "address, ahmadmalik1376@gmail.com") · GitHub · LinkedIn
- Info line: Rawalpindi, Pakistan · it's <HH:MM PKT> here

Footer

> **Wordmark:** ahmad.malik
>
> **Tagline:** Full-Stack AI Engineer in Rawalpindi, Pakistan. Drawn by hand, built with Next.js.
>
> **Icon links (accessible names):** GitHub · LinkedIn · Email
>
> **Footer navigation heading:** on this page
>
> **Footer links:** About · What I do · Work · Skills · Experience · Contact
>
> **Button:** Back to top
>
> **Copyright notice:** © 2026 Muhammad Ahmad Malik. All rights reserved. (the year is generated from the current date)
>
> **Legal disclaimers / privacy / terms:** **NOT FOUND**

404 page

- 404
- This page got erased.
- Whatever was drawn here has been rubbed out — or it was never sketched in the first place.
- Button: Back home

10. ANY OTHER TEXT

Browser/SEO metadata (src/app/layout.tsx)

> **Page title (default):** Muhammad Ahmad Malik — Full-Stack AI Engineer

> **Title template for sub-pages:** %s — M. Ahmad Malik

> **Meta description:** Muhammad Ahmad Malik, Full-Stack AI Engineer in Rawalpindi, Pakistan. I build production web apps and LLM-powered systems, and make complex AI feel simple to use.

> **Author:** Muhammad Ahmad Malik (https://github.com/AHMADMALIK1376)

> **Canonical:** /

> **Open Graph:** type website · siteName Muhammad Ahmad Malik · title/description as above

> **Twitter card:** summary_large_image

> **Theme colour:** #eaebe6

> **Language:** en

Shared-link picture (opengraph-image) — alt and text

> **Alt:** Muhammad Ahmad Malik — Full-Stack AI Engineer

> **Text printed on the image:** hello, world — i'm ;; Muhammad ;; Ahmad Malik ;; > Full-Stack AI Engineer ;; Rawalpindi, Pakistan ;; github.com/AHMADMALIK1376

Structured data for search engines (JSON-LD, schema.org Person)

- name: Muhammad Ahmad Malik
- alternateName: M. Ahmad Malik
- jobTitle: Full-Stack AI Engineer
- email: mailto:ahmadmalik1376@gmail.com
- sameAs: https://github.com/AHMADMALIK1376 ; https://www.linkedin.com/in/ahmadmalik1376/
- address: Rawalpindi, PK
- alumniOf: Iqra University, Islamabad (H-9)

robots / sitemap

> **robots.txt:** User-agent: * Allow: / Disallow: /api/ (+ sitemap and host from the site URL)

> **sitemap:** single URL (site root), changeFrequency monthly, priority 1

Contact form error messages shown to the visitor

- invalid: Something on the card doesn't look right — check the email address, and write at least a sentence.
- busy: That's a lot of cards in a short time. Please try again in a little while.
- unavailable: The post isn't connected yet. Please email me directly at ahmadmalik1376@gmail.com.
- failed: The card couldn't be sent just now. Please email me directly at ahmadmalik1376@gmail.com.

Contact API (src/app/api/contact/route.ts) — server-side text

- Email sent to ahmadmalik1376@gmail.com through Resend; sender default: Portfolio <onboarding@resend.dev>
- Subject: Portfolio message from <name>
- Body: <message> then "— <name> <<email>>" then "Sent from the contact form on the portfolio."
- Limits: name ≤ 100, email ≤ 200, message 10–5000 characters; 5 messages per visitor per hour; only this site's own pages may post; a hidden honeypot field (hp_trap_x7, label "Leave this empty") silently discards

bots.

- Error codes returned: forbidden (403), busy (429), invalid (400/413), unavailable (503), failed (502).
- Server log messages: "contact: not sent — RESEND_API_KEY is not set (put it in .env.local, and in the Vercel project's environment variables)" ; "contact: not sent — <reason>"

Illustration text (labels drawn inside the animated SVG drawings)

What I do pop-up drawings — LLM training: loss · epoch 01 · loss value · — train · -- val · lr 3e-4 · 1.2M tokens · ckpt-01.pt · layers embed, attn, mlp, head. Software architecture: floors data, services, gateway, clients; nodes browser, phone, terminal, gateway, orders, users, cdn, bus, cache; "auth · rate limit", "1.3k req/s", "12,408 rows", "hit 94%", "event bus". Web applications: sketchbook · $24 · cart 0 · → prod · up 99.98% · live · v1.0.0. Backend engineering: → api-1 · user:42 (also cart:7, sku:118, user:9) · health · req/s · cache hit · p95 · ● all systems normal · load balancer, api-1, api-2, workers, cache, postgres. Frontend development: inspect · h1.hero · button.cta · ul.cards · "margin 8 · border 1" · "padding 6 × 14" · "gap 14 · 3 columns" · "every gap equal ✓" · padding: 6px 14px; · gap: 14px; · a11y · Sign up · you@mail.com · 7.4:1.

Project pop-up drawings — Folium: you / team · edit · view · v12 · .docx. Bid & Proposal Engine: PDF · DOCX · due 14 oct · 0:00 / 60s · PASS / GAP / FAIL · NO · GO · .docx · stages intake, claude, matcher, dash, drafter. HireMe Agent: name, skills, 5 yrs · N% match · 0/3 waiting. FocusFlow: cron · JWT · 17 tables. GraphForge: 69 types · stage · no upload · GDPR ✓ · .png · .csv · engines d3, recharts, chartjs · bars, line, pie. Animated GitHub Profile: @AHMADMALIK1376 · ts, py, js · +go. NeuraCache: sara · alibaba cloud · likes: LangGraph · last: memory · chat · hi, it's sara · welcome back, Sara! still on LangGraph?. Routine Dashboard: done · this week · today · theme=light · synced ✓. Bugistan: bugistan — main.ts · ⚠ 5 problems · squashed 0 · 60 fps · canvas · 38 kinds.

Experience map illustrations: START · 2023 · NOW · HACK 2026 · git.

Accessibility text (screen-reader only / aria)

- Skip to content
- Message, at least 10 characters
- Copy email address, ahmadmalik1376@gmail.com
- Previous / Next / Previous project / Next project
- Kinds of tool
- Built with
- Main (nav aria-label)
- Footer (nav aria-label)
- Menu (dialog)
- What I do (list)
- Drive to <stop title>
- Write to Ahmad (form)

Placeholder / empty-state text

- Tell me about the project, the role, or the idea…
- your name
- you@example.com
- stamp here
- … (signature placeholder)
- a friend (envelope 'from' when no name typed)
- PKT (shown before the clock loads)

Typography / design statements in the source

Courier New is the site font (Courier Prime from Google Fonts as fallback). Palette: sage green, concrete grey, burnt sienna. Buttons are one hand-drawn design used site-wide.

Text that exists only outside the code

The downloadable resume (public/M-Ahmad-Malik-Resume.pdf, 120 KB) is a binary PDF whose text is not part of the code and is not reproduced here. The project photo is src/assets/ahmad.png (589×595).

README (repository documentation)

The repository README.md documents the design and each feature (factory, about spec sheet, scroll reveal, skills table, experience map, contact postcard, logo, design system). It is documentation, not site content, and is not reproduced here.